MariaDB CRUD Exercise Documentation: Healthcare Scheduling System

This documentation provides a comprehensive guide to completing the MariaDB CRUD exercise for a simple healthcare scheduling system. The exercise involves managing a database named "hospital" with a table called "scheduler" to store appointment information. We will cover each step in detail, including the SQL statements required, explanations of the syntax and purpose, potential execution notes, and expected outputs where applicable.

The exercise practices fundamental CRUD operations (Create, Read, Update, Delete) in MariaDB, which is a relational database management system compatible with MySQL syntax. All operations assume you are working within the "hospital" database.

Prerequisites

- **MariaDB Installation**: We've installed MariaDB on Rocky Linux 10.
- **Database Access**: We're able to log in to the MariaDB client using a tool like the MariaDB command-line client (mysql command) and phpMyAdmin (which we already installed). We use credentials with sufficient privileges (a dedicated user "sims").
- **Database Creation**: Since the "hospital" database does not exist, we create it first with following cmds:

CREATE DATABASE hospital;
USE hospital;

```
MariaDB [(none)]> CREATE DATABASE hospital;
Query OK, 1 row affected (0.009 sec)

MariaDB [(none)]> USE hospital;
Database changed
MariaDB [hospital]>
```

- **Execution Environment**: We'll run SQL statements sequentially in the MariaDB client. *Note: If using a script, save it as a .sql file and execute it with mariadb -u username -p hospital < script.sql.*

- **Error Handling**: We'll ensure no existing "**scheduler**" table exists (drop it if necessary: DROP TABLE IF EXISTS scheduler;). We'll handle any syntax errors by checking data types, constraints, and quoting.
- **Date and Time Formats**: We'll use **YYYY-MM-DD** for dates and **HH:MM:SS** for times, as specified.

Step 1: Table Creation

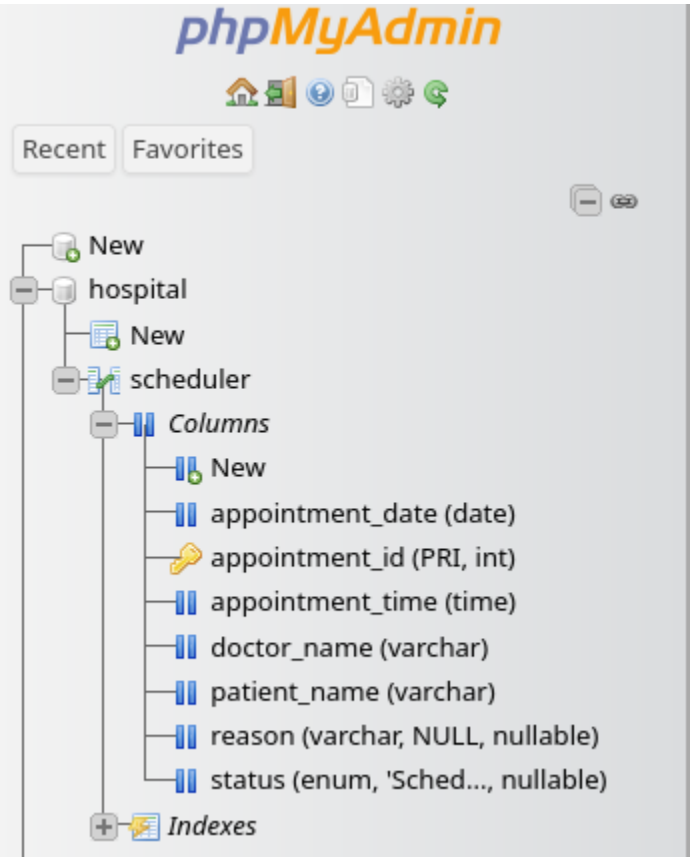
Objective

We create the "scheduler" table to store appointment details. The table includes columns for **unique appointment IDs**, **patient name**, **doctor name**, appt **date**, appt **time**, **reason** (optional), and **status** (with predefined values).

SQL Statement

```
CREATE TABLE scheduler (
  appointment_id INT PRIMARY KEY AUTO_INCREMENT NOT NULL,
  patient_name VARCHAR(100) NOT NULL,
  doctor_name VARCHAR(100) NOT NULL,
  appointment_date DATE NOT NULL,
  appointment_time TIME NOT NULL,
  reason VARCHAR(255),
  status ENUM('Scheduled', 'Completed', 'Cancelled') DEFAULT 'Scheduled'
);
```

```
MariaDB [hospital]> CREATE TABLE scheduler (
    -> appointment_id INT PRIMARY KEY AUTO_INCREMENT NOT NULL,
    -> patient_name VARCHAR(100) NOT NULL,
    -> doctor_name VARCHAR(100) NOT NULL,
    -> appointment_date DATE NOT NULL,
    -> appointment_time TIME NOT NULL,
    -> reason VARCHAR(255),
    -> status ENUM('Scheduled', 'Completed', 'Cancelled') DEFAULT 'Scheduled'
    -> );
Query OK, 0 rows affected (0.050 sec)
```



Explanation

- **appointment_id**: Serves as the primary key. AUTO_INCREMENT automatically assigns a unique integer starting from 1. NOT NULL ensures it cannot be empty.
- **patient_name and doctor_name**: Variable-length strings up to 100 characters, required (NOT NULL).
- **appointment_date and appointment_time**: Store the date (YYYY-MM-DD) and time (HH:MM:SS) separately for easy querying.
- **reason**: Optional string up to 255 characters (no NOT NULL constraint).
- **status**: Uses ENUM to restrict values to 'Scheduled', 'Completed', or 'Cancelled'. Defaults to 'Scheduled' if not specified.
- **Primary Key Constraint**: Ensures appointment_id is unique and indexed for fast lookups.

Execution Notes

- We run this in the MariaDB client after selecting the "hospital" database.

- To verify creation: SHOW TABLES; should list "**scheduler**". Describe the table: DESCRIBE scheduler; to confirm structure.

```
MariaDB [hospital]> SHOW TABLES;
+--------------------+
| Tables_in_hospital |
+--------------------+
| scheduler          |
+--------------------+
1 row in set (0.002 sec)

MariaDB [hospital]> DESC scheduler;
+------------------+----------------------------------------+------+-----+-----------+----------------+
| Field            | Type                                   | Null | Key | Default   | Extra          |
+------------------+----------------------------------------+------+-----+-----------+----------------+
| appointment_id   | int(11)                                | NO   | PRI | NULL      | auto_increment |
| patient_name     | varchar(100)                           | NO   |     | NULL      |                |
| doctor_name      | varchar(100)                           | NO   |     | NULL      |                |
| appointment_date | date                                   | NO   |     | NULL      |                |
| appointment_time | time                                   | NO   |     | NULL      |                |
| reason           | varchar(255)                           | YES  |     | NULL      |                |
| status           | enum('Scheduled','Completed','Cancelled') | YES  |     | Scheduled |                |
+------------------+----------------------------------------+------+-----+-----------+----------------+
7 rows in set (0.001 sec)
```

- Expected Output: No rows returned; success message like "Query OK, 0 rows affected".

Step 2: Inserting 10 Rows (Create Operation)

Objective

Populate the table with 10 initial appointments using the provided data. This demonstrates the Create aspect of CRUD.

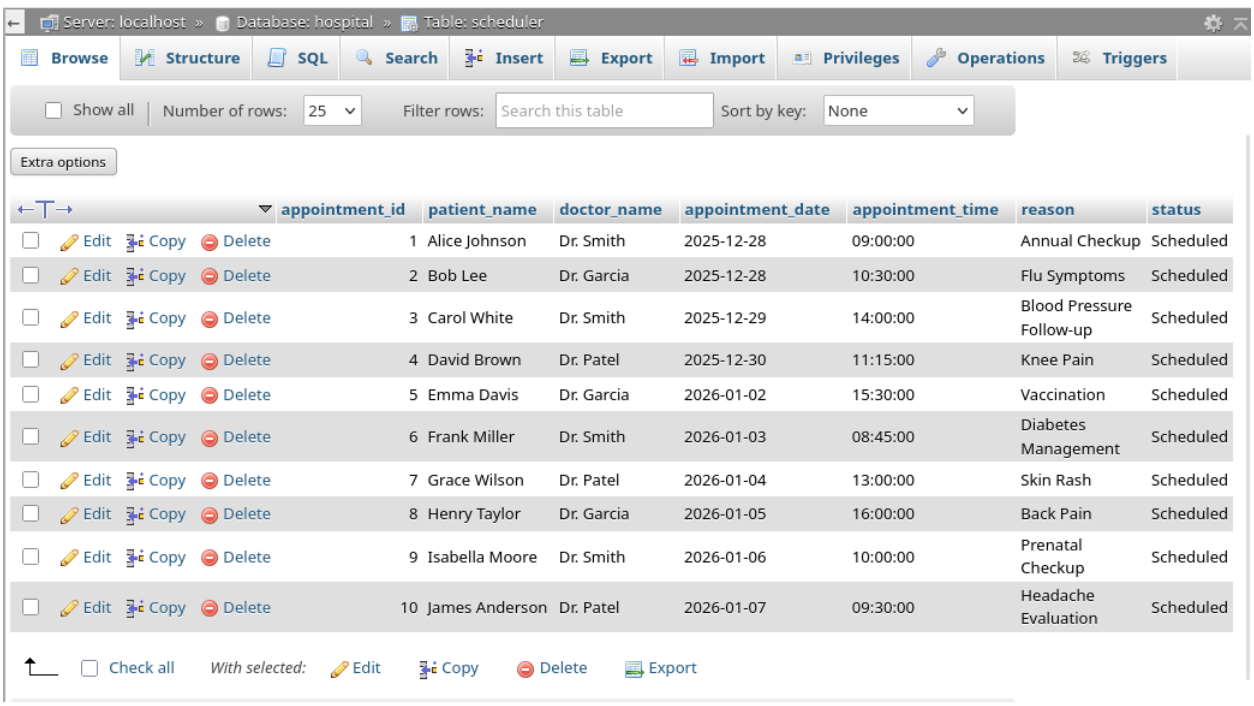
SQL Statement

We can use a single multi-row INSERT for efficiency:

```
INSERT INTO scheduler (patient_name, doctor_name, appointment_date, appointment_time, reason, status)
VALUES
  ('Alice Johnson', 'Dr. Smith', '2025-12-28', '09:00:00', 'Annual Checkup', 'Scheduled'),
  ('Bob Lee', 'Dr. Garcia', '2025-12-28', '10:30:00', 'Flu Symptoms', 'Scheduled'),
  ('Carol White', 'Dr. Smith', '2025-12-29', '14:00:00', 'Blood Pressure Follow-up', 'Scheduled'),
  ('David Brown', 'Dr. Patel', '2025-12-30', '11:15:00', 'Knee Pain', 'Scheduled'),
  ('Emma Davis', 'Dr. Garcia', '2026-01-02', '15:30:00', 'Vaccination', 'Scheduled'),
  ('Frank Miller', 'Dr. Smith', '2026-01-03', '08:45:00', 'Diabetes Management', 'Scheduled'),
  ('Grace Wilson', 'Dr. Patel', '2026-01-04', '13:00:00', 'Skin Rash', 'Scheduled'),
```

('Henry Taylor', 'Dr. Garcia', '2026-01-05', '16:00:00', 'Back Pain', 'Scheduled'),
('Isabella Moore', 'Dr. Smith', '2026-01-06', '10:00:00', 'Prenatal Checkup', 'Scheduled'),
('James Anderson', 'Dr. Patel', '2026-01-07', '09:30:00', 'Headache Evaluation', 'Scheduled');

```
MariaDB [hospital]> INSERT INTO scheduler (patient_name, doctor_name, appointment_date, appointment_time, reason, status)
    -> VALUES
    ->     ('Alice Johnson', 'Dr. Smith', '2025-12-28', '09:00:00', 'Annual Checkup', 'Scheduled'),
    ->     ('Bob Lee', 'Dr. Garcia', '2025-12-28', '10:30:00', 'Flu Symptoms', 'Scheduled'),
    ->     ('Carol White', 'Dr. Smith', '2025-12-29', '14:00:00', 'Blood Pressure Follow-up', 'Scheduled'),
    ->     ('David Brown', 'Dr. Patel', '2025-12-30', '11:15:00', 'Knee Pain', 'Scheduled'),
    ->     ('Emma Davis', 'Dr. Garcia', '2026-01-02', '15:30:00', 'Vaccination', 'Scheduled'),
    ->     ('Frank Miller', 'Dr. Smith', '2026-01-03', '08:45:00', 'Diabetes Management', 'Scheduled'),
    ->     ('Grace Wilson', 'Dr. Patel', '2026-01-04', '13:00:00', 'Skin Rash', 'Scheduled'),
    ->     ('Henry Taylor', 'Dr. Garcia', '2026-01-05', '16:00:00', 'Back Pain', 'Scheduled'),
    ->     ('Isabella Moore', 'Dr. Smith', '2026-01-06', '10:00:00', 'Prenatal Checkup', 'Scheduled'),
    ->     ('James Anderson', 'Dr. Patel', '2026-01-07', '09:30:00', 'Headache Evaluation', 'Scheduled');
Query OK, 10 rows affected (0.022 sec)
Records: 10  Duplicates: 0  Warnings: 0
```



appointment_id	patient_name	doctor_name	appointment_date	appointment_time	reason	status
1	Alice Johnson	Dr. Smith	2025-12-28	09:00:00	Annual Checkup	Scheduled
2	Bob Lee	Dr. Garcia	2025-12-28	10:30:00	Flu Symptoms	Scheduled
3	Carol White	Dr. Smith	2025-12-29	14:00:00	Blood Pressure Follow-up	Scheduled
4	David Brown	Dr. Patel	2025-12-30	11:15:00	Knee Pain	Scheduled
5	Emma Davis	Dr. Garcia	2026-01-02	15:30:00	Vaccination	Scheduled
6	Frank Miller	Dr. Smith	2026-01-03	08:45:00	Diabetes Management	Scheduled
7	Grace Wilson	Dr. Patel	2026-01-04	13:00:00	Skin Rash	Scheduled
8	Henry Taylor	Dr. Garcia	2026-01-05	16:00:00	Back Pain	Scheduled
9	Isabella Moore	Dr. Smith	2026-01-06	10:00:00	Prenatal Checkup	Scheduled
10	James Anderson	Dr. Patel	2026-01-07	09:30:00	Headache Evaluation	Scheduled

Alternatively, we could use multiple single-row INSERTs (e.g., one per row) for simplicity:

INSERT INTO scheduler (patient_name, doctor_name, appointment_date, appointment_time, reason, status)

VALUES ('Alice Johnson', 'Dr. Smith', '2025-12-28', '09:00:00', 'Annual Checkup', 'Scheduled');

-- Repeat for each of the remaining 9 rows…

Explanation

- **Columns Specified**: Explicitly lists columns to insert into, allowing flexibility if the table structure changes.
- **VALUES Clause**: Provides data in the order of the columns. Strings are single-quoted; dates and times are quoted as strings but parsed accordingly.
- **AUTO_INCREMENT**: appointment_id is omitted; MariaDB auto-assigns values (1 through 10).
- **Default Usage**: Status is explicitly set but could rely on the default if omitted.

Execution Notes

- After insertion, query SELECT COUNT(*) FROM scheduler; to confirm 10 rows.
- Expected Output: Success message like "Query OK, 10 rows affected".

```
MariaDB [hospital]> SELECT COUNT(*) FROM scheduler;
+----------+
| COUNT(*) |
+----------+
|       10 |
+----------+
1 row in set (0.001 sec)
```

<u>Step 3:</u> Read Operations

Objective

We'll retrieve data using **SELECT** queries with filters, joins (not needed here), and ordering. This covers the Read aspect of CRUD.

SQL Statements and Explanations

1. **Display all appointments scheduled with Dr. Smith.**

SELECT * FROM scheduler WHERE doctor_name = 'Dr. Smith';

- **Explanation**: Filters rows where doctor_name matches exactly. * selects all columns.
- **Expected Output** (4 rows):

```
MariaDB [hospital]> SELECT * FROM scheduler WHERE doctor_name='Dr. Smith';
+----------------+----------------+-------------+------------------+------------------+--------------------------+-----------+
| appointment_id | patient_name   | doctor_name | appointment_date | appointment_time | reason                   | status    |
+----------------+----------------+-------------+------------------+------------------+--------------------------+-----------+
|              1 | Alice Johnson  | Dr. Smith   | 2025-12-28       | 09:00:00         | Annual Checkup           | Scheduled |
|              3 | Carol White    | Dr. Smith   | 2025-12-29       | 14:00:00         | Blood Pressure Follow-up | Scheduled |
|              6 | Frank Miller   | Dr. Smith   | 2026-01-03       | 08:45:00         | Diabetes Management      | Scheduled |
|              9 | Isabella Moore | Dr. Smith   | 2026-01-06       | 10:00:00         | Prenatal Checkup         | Scheduled |
+----------------+----------------+-------------+------------------+------------------+--------------------------+-----------+
4 rows in set (0.001 sec)
```

2. **Show all appointments for December 2025, ordered by date and time.**

SELECT * FROM scheduler
WHERE appointment_date BETWEEN '2025-12-01' AND '2025-12-31'
ORDER BY appointment_date ASC, appointment_time ASC;

- **Explanation**: Uses BETWEEN for date range. ORDER BY sorts first by date, then time (ascending).
- **Expected Output** (4 rows, sorted):

```
MariaDB [hospital]> SELECT * FROM scheduler WHERE appointment_date BETWEEN '2025-12-01' AND '2025-12-31' ORDER BY appointment_date ASC, appointment_time ASC;

+----------------+----------------+-------------+------------------+------------------+--------------------------+-----------+
| appointment_id | patient_name   | doctor_name | appointment_date | appointment_time | reason                   | status    |
+----------------+----------------+-------------+------------------+------------------+--------------------------+-----------+
|              1 | Alice Johnson  | Dr. Smith   | 2025-12-28       | 09:00:00         | Annual Checkup           | Scheduled |
|              2 | Bob Lee        | Dr. Garcia  | 2025-12-28       | 10:30:00         | Flu Symptoms             | Scheduled |
|              3 | Carol White    | Dr. Smith   | 2025-12-29       | 14:00:00         | Blood Pressure Follow-up | Scheduled |
|              4 | David Brown    | Dr. Patel   | 2025-12-30       | 11:15:00         | Knee Pain                | Scheduled |
+----------------+----------------+-------------+------------------+------------------+--------------------------+-----------+
4 rows in set (0.002 sec)
```

3. **List all appointments with status 'Scheduled' along with patient name and doctor name.**

SELECT patient_name, doctor_name FROM scheduler WHERE status = 'Scheduled';

- **Explanation**: Selects only specified columns. Filters by exact status match.
- **Expected Output** (10 rows, all initially Scheduled):

```
MariaDB [hospital]> SELECT patient_name, doctor_name FROM scheduler WHERE status='Scheduled';
+----------------+-------------+
| patient_name   | doctor_name |
+----------------+-------------+
| Alice Johnson  | Dr. Smith   |
| Bob Lee        | Dr. Garcia  |
| Carol White    | Dr. Smith   |
| David Brown    | Dr. Patel   |
| Emma Davis     | Dr. Garcia  |
| Frank Miller   | Dr. Smith   |
| Grace Wilson   | Dr. Patel   |
| Henry Taylor   | Dr. Garcia  |
| Isabella Moore | Dr. Smith   |
| James Anderson | Dr. Patel   |
+----------------+-------------+
10 rows in set (0.001 sec)
```

4. **Find the appointment details for patient "David Brown".**

SELECT * FROM scheduler WHERE patient_name = 'David Brown';

- **Explanation**: Exact match on patient_name. Returns all columns for the matching row.
- **Expected Output** (1 row):

```
MariaDB [hospital]> SELECT * FROM scheduler WHERE patient_name='David Brown';
+----------------+--------------+-------------+------------------+------------------+-----------+-----------+
| appointment_id | patient_name | doctor_name | appointment_date | appointment_time | reason    | status    |
+----------------+--------------+-------------+------------------+------------------+-----------+-----------+
|              4 | David Brown  | Dr. Patel   | 2025-12-30       | 11:15:00         | Knee Pain | Scheduled |
+----------------+--------------+-------------+------------------+------------------+-----------+-----------+
1 row in set (0.001 sec)
```

<u>Step 4:</u> Update Operation

Objective

Modify existing records. This demonstrates the Update aspect of CRUD.

SQL Statements

1. **Update Alice Johnson's status to 'Completed'.**

UPDATE scheduler
SET status = 'Completed'
WHERE patient_name = 'Alice Johnson';

- **Explanation**: SET changes the status. WHERE targets the specific patient (assumes unique names; use appointment_id for precision in production).

```
MariaDB [hospital]> UPDATE scheduler SET status='Completed' WHERE patient_name='Alice Johnson';
Query OK, 1 row affected (0.009 sec)
Rows matched: 1  Changed: 1  Warnings: 0
```

2. **Reschedule Grace Wilson's appointment.**

UPDATE scheduler
SET appointment_date = '2026-01-10', appointment_time = '11:00:00'
WHERE patient_name = 'Grace Wilson';

- **Explanation**: Updates multiple columns in one statement. Targets by patient name.

```
MariaDB [hospital]> UPDATE scheduler SET appointment_date='2026-01-10', appointment_time='11:00:00' WHERE patient_name='Grace Wilson';
Query OK, 1 row affected (0.003 sec)
Rows matched: 1  Changed: 1  Warnings: 0
```

Execution Notes

- Confirm changes: Run SELECT * FROM scheduler WHERE patient_name IN ('Alice Johnson', 'Grace Wilson');.
- Expected Output: "Query OK, 1 row affected" for each.

<u>Step 5:</u> Delete Operation

Objective

We'll remove a record. This covers the **Delete** aspect of CRUD.

SQL Statement

DELETE FROM scheduler WHERE patient_name = 'Bob Lee';

```
MariaDB [hospital]> DELETE FROM scheduler WHERE patient_name='Bob Lee';
Query OK, 1 row affected (0.011 sec)
```

Explanation

- **DELETE FROM**: Specifies the table.
- **WHERE**: Targets the row by patient name. Without WHERE, all rows would be deleted (dangerous!).

Execution Notes

- Confirm: SELECT COUNT(*) FROM scheduler; should return 9.

- Expected Output: "Query OK, 1 row affected".

Final Task

Objective

We'll verify the table's state after all operations.

SQL Statement

SELECT * FROM scheduler ORDER BY appointment_date, appointment_time;

```
MariaDB [hospital]> SELECT * FROM scheduler ORDER BY appointment_date, appointment_time;
+----------------+----------------+-------------+------------------+------------------+--------------------------+-----------+
| appointment_id | patient_name   | doctor_name | appointment_date | appointment_time | reason                   | status    |
+----------------+----------------+-------------+------------------+------------------+--------------------------+-----------+
|              1 | Alice Johnson  | Dr. Smith   | 2025-12-28       | 09:00:00         | Annual Checkup           | Completed |
|              3 | Carol White    | Dr. Smith   | 2025-12-29       | 14:00:00         | Blood Pressure Follow-up | Scheduled |
|              4 | David Brown    | Dr. Patel   | 2025-12-30       | 11:15:00         | Knee Pain                | Scheduled |
|              5 | Emma Davis     | Dr. Garcia  | 2026-01-02       | 15:30:00         | Vaccination              | Scheduled |
|              6 | Frank Miller   | Dr. Smith   | 2026-01-03       | 08:45:00         | Diabetes Management      | Scheduled |
|              8 | Henry Taylor   | Dr. Garcia  | 2026-01-05       | 16:00:00         | Back Pain                | Scheduled |
|              9 | Isabella Moore | Dr. Smith   | 2026-01-06       | 10:00:00         | Prenatal Checkup         | Scheduled |
|             10 | James Anderson | Dr. Patel   | 2026-01-07       | 09:30:00         | Headache Evaluation      | Scheduled |
|              7 | Grace Wilson   | Dr. Patel   | 2026-01-10       | 11:00:00         | Skin Rash                | Scheduled |
+----------------+----------------+-------------+------------------+------------------+--------------------------+-----------+
9 rows in set (0.001 sec)
```

Explanation

- Selects all columns and rows.
- Orders by date, then time (ascending) for chronological view.

Expected Final State

After insertions (10 rows), updates (2 changes), and deletion (1 removal), 9 rows remain. Alice's status is 'Completed'; Grace's date/time updated; Bob deleted.